4/2/19 Tables, squares, cubes, power lists, last digit

12X1=12	13X1=13	14X1=14	15X1=15	16X1=16
12X2=24	13X2=26	14X2=28	15X2=30	16X2=32
12X3=36	13X3=39	14X3=42	15X3=45	16X3=48
12X4=48	13X4=52	14X4=56	15X4=60	16X4=64
12X5=60	13X5=65	14X5=70	15X5=75	16X5=80
12X6=72	13X6=78	14X6=84	15X6=90	16X6=96
12X7=84	13X7=91	14X7=98	15X7=105	16X7=112
12X8=96	13X8=104	14X8=112	15X8=120	16X8=128
12X9=108	13X9=117	14X9=126	15X9=135	16X9=144
12X10=120	13X10=130	14X10=140	15X10=150	16X10=160

17X1=17	18X1=18	19X1=19
17X2=34	18X2=36	19X2=38
17X3=51	18X3=54	19X3=57
17X4=68	18X4=72	19X4=76
17X5=85	18X5=90	19X5=95
17X6=102	18X6=108	19X6=114
17X7=119	18X7=126	19X7=133
17X8=136	18X8=144	19X8=152
17X9=153	18X9=162	19X9=171
17X10=170	18X10=180	19X10=190

Last digit / Power	0	1	2	3	4
1	1	1	1	1	1
2	1	2	4	8	6
3	1	3	9	7	1
4	1	4	6	4	6
5	1	5	5	5	5
6	1	6	6	6	6
7	1	7	9	3	1
8	1	8	4	2	6
9	1	9	1	9	1
0	n.d.	0	0	0	0

Power list ⇒

$2^0 = 1$	$3^0 = 1$
$2^1 = 2$	$3^1 = 3$
$2^2 = 4$	$3^2 = 9$
$2^3 = 8$	$3^3 = 27$
$2^4 = 16$	$3^4 = 81$
$2^5 = 32$	$3^5 = 243$
$2^6 = 64$	★ $3^6 = 729 = 27^2 = 9^3$
$2^7 = 128$	
$2^8 = 256$	
$2^9 = 512$	
$32^2 = 2^{10} = 1024$	

$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Square list ⇒

$1^2 = 1$	$11^2 = 121$	$21^2 = 441$	$31^2 = 961$
$2^2 = 4$	$12^2 = 144$	$22^2 = 484$	$32^2 = 1024$
$3^2 = 9$	$13^2 = 169$	$23^2 = 529$	$33^2 = 1089$
$4^2 = 16$	$14^2 = 196$	$24^2 = 576$	$34^2 = 1156$
$5^2 = 25$	$15^2 = 225$	$25^2 = 625$	$35^2 = 1225$
$6^2 = 36$	$16^2 = 256$	$26^2 = 676$	$36^2 = 1296$
$7^2 = 49$	$17^2 = 289$	$27^2 = 729$	$37^2 = 1369$
$8^2 = 64$	$18^2 = 324$	$28^2 = 784$	$38^2 = 1444$
$9^2 = 81$	$19^2 = 361$	$29^2 = 841$	$39^2 = 1521$
$10^2 = 100$	$20^2 = 400$	$30^2 = 900$	$40^2 = 1600$

Cubes

$1^3 = 1$	$6^3 = 216$	$11^3 = 1331$	$16^3 = 4096 = 2^{12}$
$2^3 = 8$	$7^3 = 343$	$12^3 = 1728$	$17^3 = 4913$
$3^3 = 27$	$8^3 = 512$	$13^3 = 2197$	$18^3 = 5832$
$4^3 = 64$	$9^3 = 729$	$14^3 = 2744$	$19^3 = 6859$
$5^3 = 125$	$10^3 = 1000$	$15^3 = 3375$	$20^3 = 8000$

4/2/19

Binomial Theorem

- $(a+x)^n = {}^nC_0 a^n x^0 + {}^nC_1 a^{n-1} x^1 + {}^nC_2 a^{n-2} x^2 + \dots + {}^nC_n a^0 x^{n-1}$

, $a \in R$, no of terms $= n+1$

- $(x+a)^n + (x-a)^n = 2\{{}^nC_0 x^n + {}^nC_2 x^{n-2} a^2 + {}^nC_4 x^{n-4} a^4 + \dots\dots\dots$

no. of terms $= \frac{n}{2}+1$, n = even or $= \frac{n+1}{2}$, n = odd

- $(x+a)^n - (x-a)^n = 2\{{}^nC_1 x^{n-1} a^1 + {}^nC_3 x^{n-3} a^3 + {}^nC_5 x^{n-5} a^5 + \dots\dots\dots$

no. of terms $= \begin{cases} \frac{n}{2}, & n \in \text{even} \\ \frac{n+1}{2}, & n \in \text{odd} \end{cases}$

- **General Term** – of $(x+a)^n$

$T_{r+1} = {}^nC_r x^{n-r} \cdot a^r$

- **Middle Term** – $(x+a)^n$

i) $n \in$ even $\Rightarrow$ $T_{\frac{n}{2}+1} = {}^nC_{n/2} x^{n/2} a^{n/2}$

ii) $n \in$ odd $\Rightarrow$ ① $T_{\frac{n+1}{2}} = {}^nC_{\frac{n-1}{2}} \cdot x^{\frac{n+1}{2}} \cdot a^{\frac{n-1}{2}}$

② $T_{\frac{n-1}{2}} = {}^nC_{\frac{n+1}{2}} \cdot x^{\frac{n-1}{2}} \cdot a^{\frac{n+1}{2}}$

- **Greatest coefficient**

i) if n is even $\Rightarrow {}^nC_{n/2}$

ii) if n is odd $\Rightarrow {}^nC_{\frac{n-1}{2}}$ and ${}^nC_{\frac{n+1}{2}}$

- **Some props -**

1) The coefficient of terms equidistant from the begining and the end are equal

2) The sum of binomial coefficients in the expansion is $(1+x)^n = 2^n$

$$2^n = {}^nC_0 + {}^nC_1 + \dots + {}^nC_n$$

3) Sum of odd coeffi = sum of even coeffi

$$C_1 + C_3 + \dots = C_0 + C_2 + C_4 + \dots = 2^{n-1}$$

4) In expansion of $(x - 2y + 3z)^n$, put $x = y = z = 1$ to get sum of coefficients $= (1 - 2 + 3)^n = 2^n$

5) In expansion of $(1 + x + x^2)^n$ put ~~x~~ $x = 1$, to get sum of coefficients $= (1+1+1)^n = 3^n$

- $x^n - a^n = (x-a)(x^{n-1} + x^{n-2}\cdot a^1 + x^{n-3}\cdot a^2 + \dots + a^{n-1})$, $\forall n \in N$

- $x^n + a^n = (x+a)(x^{n-1} - x^{n-2}\cdot a^1 + x^{n-3}\cdot a^2 + \dots + x^{n-r}\cdot a^{r-1})$ $\forall$ $n \in$ odd

- $2 \leq \left(1 + \frac{1}{n}\right)^n < 3$, $n \geq 1$, $n \in N$

- $\left(\frac{n}{3}\right)^n < n! < \left(\frac{n}{2}\right)^n$, $n > 6$

- **Multinomial Theorem-**

If n is +ve and $x_1, x_2, x_3, \ldots\ldots, x_k \in \mathbb{C}$, then

$$(x_1 + x_2 + x_3 + \ldots\ldots + x_k)^n =$$

$$\sum \frac{n!}{(\alpha_1!)(\alpha_2!)(\alpha_3!)\ldots\ldots(\alpha_k!)} x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdot x_3^{\alpha_3} \ldots\ldots x_k^{\alpha_k}$$

where $\alpha_1, \alpha_2, \ldots\ldots, \alpha_k$ are non-ve and $\sum_{i=1}^{k} \alpha_i = n$

- The coefficient of $x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdot x_3^{\alpha_3} \ldots\ldots x_k^{\alpha_k}$ is

$$\sum \frac{n!}{(\alpha_1!)(\alpha_2!)(\alpha_3!)\ldots\ldots(\alpha_k!)}$$

- $(a+b+c)^n = \sum \frac{n!}{(\alpha!)(\beta!)(\gamma!)} a^{\alpha} b^{\beta} c^{\gamma}$

$\alpha + \beta + \gamma = n$

- The no. of distinct terms in multinomial expansion of $(x_1 + x_2 + x_3 + \ldots\ldots + x_k)^n$ is

$${}^{n+k-1}C_{k-1}$$

- ~~The~~ Greatest coefficient in the expansion of

$$(x_1 + x_2 + x_3 + \ldots\ldots + x_k)^n \text{ is } \frac{n!}{(q!)^{k-r}((q+1)!)^r}$$

where q is quotient and r is remainder when n is divided by k i.e. $k \overline{\left| n \right.} \, q$, r

- **Coefficient of x^r in the expansion**

If n is +ve and $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_k \in C$ then coefficient of x^r in expansion of

$$(a_1 + a_2 x + a_3 x^2 + \ldots + a_k x^{k-1})^n \text{ is}$$

$$\sum \frac{n!}{(\alpha_1!)(\alpha_2!)(\alpha_3!)\ldots(\alpha_k!)} a_1^{\alpha_1} \cdot a_2^{\alpha_2} \ldots a_k^{\alpha_k}$$

$$\alpha_1 + \alpha_2 + \ldots + \alpha_k = n, \quad \alpha_2 + 2\alpha_3 + 3\alpha_4 + \ldots + (k-1)\alpha_k = r$$

* **Summation formula-**

1) $\sum_{r=0}^{n} {}^nC_r \cdot x^r = (1+x)^n$

2) $\sum_{r=0}^{n} {}^nC_r (-1)^r \cdot x^r = (1-x)^n$

3) $\sum_{r=0}^{n} \binom{n}{r} = 2^n$

4) $\sum_{r=0}^{n} \binom{n}{2r} = 2^{n-1} = \sum_{r=0}^{n} \binom{n}{2r+1}$

5) $\sum_{r=1}^{n} (-1)^r r^a \cdot {}^nC_r = 0, \quad a \geq 0, \; a \in I$

6) $\sum_{r=1}^{n} r^1 \cdot {}^nC_r = n \cdot 2^{n-1}$

7) $\sum_{r=1}^{n} r^2 \cdot {}^nC_r = n(n+1) \cdot 2^{n-2}$

8) $\sum_{r=1}^{n} r^3 \cdot {}^nC_r = n^2(n+3) \cdot 2^{n-3}$

9) $\sum_{r=1}^{n} r^4 \cdot {}^nC_r = n(n^3+6n^2+3n-2) \cdot 2^{n-4}$

10) $\sum_{r=0}^{n} \frac{C_r}{r+1} = \frac{2^{n+1}-1}{n+1}$

11) $\sum_{r=0}^{n} (-1)^r \cdot \frac{C_r}{(r+1)} = \frac{1}{n+1}$

12) $\sum_{r=1}^{n} \cdot \left({}^nC_r\right)^2 = {}^{2n}C_n$

13) $\sum_{r=1}^{n} r \cdot \left({}^nC_r\right)^2 = \frac{n}{2}\left[{}^{2n}C_n\right] = n \cdot {}^{2n-1}C_{n-1}$

14) $\sum_{r=1}^{n} r^2 \cdot \left({}^nC_r\right)^2 = n^2 \cdot {}^{2(n-1)}C_{n-1}$

15) $C_0 \cdot C_r + C_1 \cdot C_{r+1} + C_2 \cdot C_{r+2} + C_3 C_{r+3} + \ldots$
$\ldots + C_{n-r} \cdot C_n = {}^{2n}C_{n-r}$ or ${}^{2n}C_{n+r}$

16) $\sum_{0 \le r \le s \le n} \sum C_r \cdot C_s = \frac{1}{2}\left(2^{2n} - {}^{2n}C_n\right)$

17) $\sum_{0 \le r \le s \le n} \sum C_r \cdot C_s = \frac{1}{2}\left(2^{2n} + {}^{2n}C_n\right)$

18) $\sum_{r \neq s} \sum C_r \cdot C_s = 2^{2n} - {}^{2n}C_n$

19) $\sum_{0 \leq r < s \leq n} \sum (C_r + C_s) = n \cdot \left(\sum_{r=0}^{n} {}^{n}C_r \right) = n \cdot 2^n$

20) $\sum_{0 \leq r < s \leq n} \sum \left(C_r^2 + C_s^2 \right) = n \cdot \left(\sum_{r=0}^{n} {}^{n}C_r^2 \right) = n \cdot {}^{2n}C_n$

21) $\sum_{r=0}^{n} (-1)^r \left({}^{n}C_r \right)^2 = \begin{cases} 0, & n \in \text{odd} \\ (-1)^{\frac{n}{2}} \cdot {}^{n}C_{\frac{n}{2}}, & n \in \text{even} \end{cases}$

22) ${}^{n}C_0 \cdot {}^{m}C_r + {}^{n}C_1 \cdot {}^{m}C_{r-1} + {}^{n}C_2 \cdot {}^{m}C_{r-2} + \ldots + {}^{n}C_n \cdot {}^{m}C_m = {}^{m+n}C_r$

23) ${}^{n}C_0 \cdot {}^{m}C_r + {}^{n}C_1 \cdot {}^{m}C_{r+1} + {}^{n}C_2 \cdot {}^{m}C_{r+2} + \ldots + {}^{n}C_n \cdot {}^{m}C_m = {}^{m+n}C_{m-r}$

24) $\sum_{r=0}^{n} (-1)^r \cdot \frac{{}^{n}C_r}{r} = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$

25) Four consecutive binomial coefficients can never be in AP and 3 consecutive binomial coefficients can never be in in G.P. or H.P.

$sA + cA = \sqrt{1 + s2A}$

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Ch- Trigonometry and SOT

* Trigo formulae-

1) $s(A \pm B) = sA\,cB \pm cA\,sB$

2) $c(A \pm B) = cA\,cB \mp sA \cdot sB$

3) $s(A+B) \cdot s(A-B) = s^2A - s^2B$

* 4) $c(A+B) \cdot c(A-B) = c^2A - s^2B = c^2B - s^2A$

5) $s(A+B) + s(A-B) = 2\,s\,A \cdot cB$

~~6) $c(A+B) + c(A-B) = 2\,cA \cdot sB$~~

6) $s(A+B) - s(A-B) = 2\,cA\,sB$

7) $c(A+B) + c(A-B) = 2\,cA \cdot cB$

* 8) $c(A+B) - c(A-B) = -2\,sA \cdot sB$

9) $sC + sD = 2\,s\left(\frac{C+D}{2}\right) \cdot c\left(\frac{C-D}{2}\right)$

10) $sC - sD = 2\,c\left(\frac{C+D}{2}\right) \cdot s\left(\frac{C-D}{2}\right)$

11) $cC + cD = 2\,c\left(\frac{C+D}{2}\right) \cdot c\left(\frac{C-D}{2}\right)$

* 12) $cC - cD = -2\,s\left(\frac{C+D}{2}\right) \cdot s\left(\frac{C-D}{2}\right)$

13) $\tan(A \pm B) = \frac{tA \pm tB}{1 \mp tA \cdot tB}$

41) $s^6 + c^6 = 1 - 3s^2c^2$

42) $s^4 + c^4 = 1 - 2s^2c^2$

★14) $\cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$

15) $s(A+B+C) = (\Sigma\, sA\, cB\, cC) + (\Pi\, sA)$

16) $c(A+B+C) = (\Pi\, cA) - (\Sigma\, cA\, sB\, sC)$

★17) $t(A+B+C) = \frac{(\Sigma\, tA) - (\Pi\, tA)}{1 - (\Sigma\, tA \cdot tB)}$

18) $t(A_1 + A_2 + A_3 + \dots) = \frac{S_1 - S_3 + S_5 - S_7 + \dots}{1 - S_2 + S_4 - S_6 + S_8 - \dots}$

S_1 = Sum of terms taken 1 @ a time

S_2 = " " " " 2 " " "

S_3 = " " " " 3 " " "

★19) $s(2A) = 2\,sA\,cA = \frac{2\,tA}{1 + t^2A}$

★20) $c(2A) = c^2A - s^2A = 2c^2A - 1 = 1 - 2s^2A = \frac{1 - t^2A}{1 + t^2A}$

★21) $t(2A) = \frac{2\,tA}{1 - t^2A}$

★22) $c^2\theta = \frac{1 + c2\theta}{2}$

★23) $s^2\theta = \frac{1 - c2\theta}{2}$

39) $s^2\theta + s^2(60+\theta) + s^2(60-\theta) = \frac{3}{2}$

40) $c\theta \cdot c2\theta \cdot c4\theta \cdot \dots \cdot c(2^{n-1}\theta) = \frac{s(2^n\theta)}{2^n s\theta}$

24) $t^2\theta = \frac{1 - c2\theta}{1 + c2\theta}$

25) $s(3\theta) = 3s\theta - 4s^3\theta$

26) $c(3\theta) = 4c^3\theta - 3c\theta$

*27) $t(3\theta) = \frac{3t\theta - t^3\theta}{1 - 3t^2\theta}$

28) $t(3\theta) \cdot t(2\theta) \cdot t(\theta) = t(3\theta) - t(2\theta) - t(\theta)$

29) $\cot\theta - \tan\theta = 2\cot(2\theta)$

30) $\cot\theta + \tan\theta = 2\operatorname{cosec}(2\theta)$

31) $s\theta \cdot s(60-\theta) \cdot s(60+\theta) = \frac{1}{4}\sin(3\theta)$

32) $c\theta \cdot c(60-\theta) \cdot c(60+\theta) = \frac{1}{4}\cos(3\theta)$

33) $t\theta \cdot t(60-\theta) \cdot t(60+\theta) = \frac{1}{4}\tan(3\theta)$

34) $s^4\theta + c^2\theta = s^2\theta + c^4\theta$

35) $1 - s\theta = \left(c\left(\frac{\theta}{2}\right) - s\left(\frac{\theta}{2}\right)\right)^2$

36) $1 + s\theta = \left(c\left(\frac{\theta}{2}\right) + s\left(\frac{\theta}{2}\right)\right)^2$

37) $\sqrt{1 - c\theta} = \sqrt{2}\left|s\left(\frac{\theta}{2}\right)\right|$

38) $\tan\frac{\theta}{2} = \frac{1 - \cos\theta}{s\theta} = \frac{s\theta}{1 + c\theta}$

8) $\sum cA = 1 + 4\left(\prod s\left(\frac{A}{2}\right)\right)$

• Conditional Identities

⇒ If $A+B+C = \pi\ (180^\circ)$

1) $\sum tA = \prod tA$

2) $\sum t\left(\frac{A}{2}\right) \cdot t\left(\frac{B}{2}\right) = 1$

3) $\sum \cot A \cdot \cot B = 1$

4) $\sum \cot\left(\frac{A}{2}\right) = \prod \cot\left(\frac{A}{2}\right)$

5) $s2A + s2B + s2C = 4sA\,sB\,sC = 4\left(\prod sA\right)$

6) $c2A + c2B + c2C = -1 - 4\left(\prod cA\right)$

7) $sA + sB + sC = 4\prod\left(\cos\left(\frac{A}{2}\right)\right)$

⇒ Summation of trigo series -

1) $s\alpha + s(\alpha+\beta) + s(\alpha+2\beta) + \dots\dots + s(\alpha+(n-1)\beta)$

$$= \frac{\sin\left(\frac{n\beta}{2}\right) \cdot \boxed{s}\left(\frac{2\alpha + (n-1)\beta}{2}\right)}{s\left(\frac{\beta}{2}\right)}$$

2) $c\alpha + c(\alpha+\beta) + c(\alpha+2\beta) + \dots\dots + c(\alpha+(n-1)\beta)$

$$= \frac{s\left(\frac{n\beta}{2}\right) \cdot \boxed{c}\left(\frac{2\alpha + (n-1)\beta}{2}\right)}{s\left(\frac{\beta}{2}\right)}$$

$\Sigma \tan A, \tan B = 1$, $\Sigma A = (2n+1)\frac{\pi}{2}$, $n \in I$

* Some standard angles -

1) $\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}} = \sin 165^\circ$

2) $\cos 15^\circ = \sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 105^\circ$

3) $\tan 15^\circ = \cot 75^\circ = 2-\sqrt{3}$

4) $\cot 15^\circ = \tan 75^\circ = 2+\sqrt{3}$

5) $\tan 22.5 = \cot 67.5 = \sqrt{2}-1 = \tan \frac{\pi}{8}$

6) $\cot 22.5 = \tan 67.5 = \sqrt{2}+1 = \cot \frac{\pi}{8}$

7) $\sin 18^\circ = \cos 72^\circ = \frac{\sqrt{5}-1}{4} = \sin \frac{\pi}{10}$

8) $\cos 36^\circ = \sin 54 = \frac{\sqrt{5}+1}{4} = \sin \frac{\pi}{5}$

X 9) $\sin 22.5 = \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}}} = \frac{1}{2}\left(\sqrt{2-\sqrt{2}}\right) = \sin \frac{\pi}{8}$

* Maximum and minimum values

1) $a\cos\theta + b\sin\theta$ will lie bt^n $\left[-\sqrt{a^2+b^2}, +\sqrt{a^2+b^2}\right]$

2) Min value of $a^2\tan^2\theta + b^2\cot^2\theta = ab$; $(a, b > 0)$

3) $-\sqrt{a^2+b^2+2ab\cos(\alpha-\beta)} \leq a\cos(\alpha+\theta) + b\cos(\beta+\theta) \leq +\sqrt{a^2+b^2+2ab\cos(\alpha-\beta)}$

* SOT

1) Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R = \frac{abc}{2\Delta}$$

$$a : b : c = \sin A : \sin B : \sin C$$

2) Cosine rule -

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad (2 \text{ more})$$

3) Projection formula -

$$a = b\cos C + c\cos B$$
$$b = c\cos A + a\cos C$$
$$c = b\cos A + a\cos B$$

4) Tangent rule / Napier's analogy -

$$\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\left(\frac{A}{2}\right)$$

5) Area of Δ (Δ - delta) -

i) $\Delta = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A = \frac{1}{2} ac \sin B$

× ii) $\Delta = \frac{1}{2} a^2 \frac{\sin C \sin B}{\sin A}$ (2 more)

iii) $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$, s = semiperimeter (heron's formula)

6) Half-angle formula

(I)

i) $\sin\frac{A}{2} = \sqrt{\frac{(s-c)(s-b)}{bc}}$

ii) $\sin\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}$

iii) $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(II) i) $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$

ii) $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$

iii) $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

(III) i) $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

ii) $\tan\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$

iii) $\tan\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$

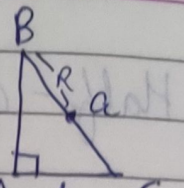
7) Circumcircle of Δ (R)-

i) $R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}$

ii) $\forall$ equilateral Δ, $R = \frac{a}{\sqrt{3}} = \frac{b}{\sqrt{3}} = \frac{c}{\sqrt{3}}$

$\therefore a^2 + b^2 + c^2 = 9R^2$ ($a = b = c = \sqrt{3}R$)

iii) $\forall$ right angle Δ $R = \frac{a}{2}$



$a^2 = b^2 + c^2$, $a^2 + b^2 + c^2 = 8R^2$

8) Incircle of Δ (r)

i) $r = \frac{\Delta}{s}$

ii) $r = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2}$

xiii) $r = a\frac{\sin\frac{B}{2}\cdot\sin\frac{C}{2}}{\cos\frac{A}{2}}$ (2 more)

iv) $r = 4R\left(\pi \sin\frac{A}{2}\right) = 4R\sin\frac{A}{2}\cdot\sin\frac{B}{2}\cdot\sin\frac{C}{2}$

v) Distance of incentre from sides is r & from vertices (angular points) is $r\operatorname{cosec}\frac{A}{2}$, $r\operatorname{cosec}\frac{B}{2}$, $r\operatorname{cosec}\frac{C}{2}$ from A, B, C.

*vi) $\forall$ eq^n Δ, $r = \frac{a}{2\sqrt{3}}$ $(a=b=c)$

$r = \frac{R}{2}$, $a^2 + b^2 + c^2 = 36r^2$

*vii) $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$

always $\Rightarrow$ $\left[\cos A + \cos B + \cos C \le \frac{3}{2} \Rightarrow r \le \frac{R}{2}\right]$

equality holds $\forall$ equilaterals.

9) Escribed circles (r_1, r_2, r_3):-

i) $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$, $r_3 = \frac{\Delta}{s-c}$

ii) $r_1 = s\tan\frac{A}{2}$, $r_2 = s\tan\frac{B}{2}$, $r_3 = s\tan\frac{C}{2}$

xiii) $r_1 = a\frac{C\frac{B}{2}\cdot C\frac{C}{2}}{C\frac{A}{2}}$ (2 more)

xiv) $r_1 = 4R\, S\frac{A}{2}\cdot C\frac{B}{2}\cdot C\frac{C}{2}$ (2 more)

v) If $r_1, r_2, r_3 \Rightarrow$ H.P. so $a, b, c \Rightarrow$ A.P.

vi) $r \cdot r_1 \cdot r_2 \cdot r_3 = \Delta^2$

x 10) Length of angle bisector

$$AD = \frac{2\Delta}{(b+c)}\operatorname{cosec}\frac{A}{2} = \frac{2bc}{(b+c)}\cos\frac{A}{2}$$

$$BE = \frac{2\Delta}{(a+c)}\operatorname{cosec}\frac{B}{2} = \frac{2ac}{(a+c)}\cos\frac{B}{2}$$

$$CF = \frac{2\Delta}{(a+b)}\operatorname{cosec}\frac{C}{2} = \frac{2ab}{a+b}\cos\frac{C}{2}$$

* 11) Length of Median

$$AD = \frac{1}{2}\sqrt{2b^2+2c^2-a^2},\quad BE = \frac{1}{2}\sqrt{2a^2+2c^2-b^2}$$

$$CF = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2}$$

*12) If l_i $(i = 1, 2, 3)$ are lengths of medians then $\left[4(l_1^2 + l_2^2 + l_3^2) = 3(a^2 + b^2 + c^2)\right]$

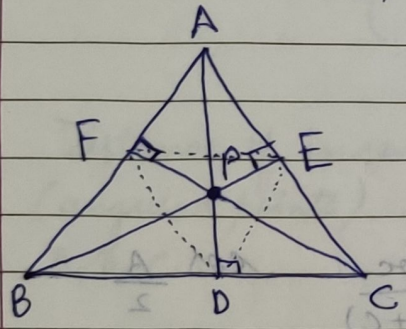
13) Distance of orthocentre (P) from vertices

$PA = 2R\cos A$ $\quad PC = 2R\cos C$

$PB = 2R\cos B$

14) Distance of orthocentre (P) from sides

$PD = 2R\cos B \cdot \cos C$, $PE = 2R\cos A \cdot \cos C$; $PF = 2R\cos A \cdot \cos B$



15) Pedal Δ $\Rightarrow$ Δ formed by the foot of the $\perp$ drawn from vertices of the Δ.

ΔDEF is pedal Δ of ΔABC

angles (i) $\angle D = \pi - 2A$, $\angle E = \pi - 2B$, $\angle F = \pi - 2C$

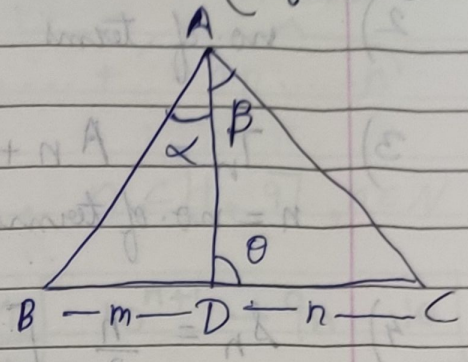
sides (ii) $DF = R\sin 2B$, $FE = R\sin 2A$, $ED = R\sin 2C$

(iii) orthocentre of a Δ (ΔABC) is incentre of its pedal Δ (ΔDEF)

(iv) $R' = \frac{R}{2}$, circumradius of ΔDEF is $\left(\frac{1}{2}\right.$ circumradius $\left.\text{ of } \Delta ABC\right)$

16) m-n theorem -

$\frac{BD}{CD} = \frac{m}{n}$ (ratio not length)



i) $(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

ii) $(m+n)\cot\theta = n\cot B - m\cot C$

17) Regular polygon of n sides -

i) Sum of all internal angle $= (n-2)\pi$, $n \geq 3$

ii) each internal angle $= \frac{(n-2)\pi}{n}$

Let side of a regular polygon $= a$

A) $R = \frac{a}{2}\operatorname{cosec}\frac{\pi}{n}$

B) $r = \frac{a}{2}\cot\frac{\pi}{n}$

c) $\frac{r}{R} = \cos\frac{\pi}{n}$

D) Area of polygon $= \frac{n}{4}a^2\cot\frac{\pi}{n}$

18) $\Delta \leq \frac{s^2}{3\sqrt{3}}$ (general)

5/2/18 Ch- Sequence and series

AP 1) $T_n = a + (n-1)\,d = l$

2) no. of terms $= n = \left[\frac{l-a}{d}\right] + 1$, $[\cdot]$ = GIF

3) $T_n = An + B$

n = no. of terms ; common difference $= A = d$

4) $S_n = \frac{n}{2}\left[2a + (n-1)\,d\right]$

5) $S_n = An^2 + Bn$, $d = 2A$

$\Rightarrow T_n = 2An + (B - A)$

6) n^{th} term from end

$t'_n = l - (n-1)\,d$

7) $T_n = \frac{T_{n-k} + T_{n+k}}{2}$

GP $a, ar, ar^2, \ldots\ldots, ar^{n-1}$

1) $T_n = ar^{n-1} = l$

2) $S_n = \frac{a(1-r^n)}{(1-r)} = \frac{a - lr}{1-r} = \frac{a - rl}{1-r}$

3) $S_\infty = \frac{a}{1-r}$, $0 < |r| < 1$

4) $T_n = \sqrt{T_{n-k} \cdot T_{n+k}}$

5) $a, b, c, \ldots \Rightarrow$ G.P.

so, $\log a, \log b, \log c \Rightarrow$ A.P.

6) $S = n + nn + nnn + \ldots\ldots + \underbrace{nnn\ldots n}_{n \text{ times}}$

~~$S = \frac{n}{81}(10^{n+1} - 9n - 10)$, $n \in [1, 9]$, $n \in N$~~

H.P. 1) $a, b, c \Rightarrow$ HP

so $\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \Rightarrow$ A.P.

$$\boxed{S = \frac{n}{81}\left[10^{n+1} - 9n - 10\right]}$$

$$\boxed{T_n = \frac{1}{a + (n-1)d}}$$

* AM, GM, HM -

(A) AM

i) single AM $\Rightarrow$ a A b

$$\boxed{A = \frac{a+b}{2}}$$

ii) n AM's $\Rightarrow$

$\boxed{a \quad A_1 \quad A_2 \quad A_3 \ldots\ldots A_n \quad b} \Rightarrow n+2$ terms in A.P.

$T_{n+2} = b$

$a + (n+1)d = b$

$$\boxed{d = \frac{b-a}{n+1}}$$

iii) Sum of n AM's $= n \cdot A = n\left(\frac{a+b}{2}\right)$

(B) GM

i) single GM ⇒ a G b

$$G = \sqrt{ab}$$

ii) n GM's ⇒ [a G_1 G_2 G_3 G_4 ..... G_n b] ⇒ n+2 terms

$$T_{n+2} = b = a r^{n+1}$$

$$\star \left[r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}} \right]$$

iii) Product of n GM's ⇒ $= G^n = \left(\sqrt{ab}\right)^n = (ab)^{\frac{n}{2}}$

(C) HM (i) single HM ⇒ a H b ⇒ H.P. ⇒ A.P.

$\frac{1}{a}$ $\frac{1}{H}$ $\frac{1}{b}$

$$\left[H = \frac{2ab}{a+b} \right]$$

ii) n HM's ⇒ a H_1 H_2 ..... H_n b

$$\left[d = \frac{a-b}{ab(n+1)} \right]$$

iii) Sum of n HM's ⇒ $= \frac{n}{H} = \frac{n(a+b)}{2ab}$

$$= \frac{n}{2}\left(\frac{1}{a} + \frac{1}{b}\right)$$

$A = A.M.$, $G = G.M.$, $H = H.M.$

a) for 2 +ve nos $\Rightarrow a, b$

$$\boxed{G^2 = A \cdot H}$$

b) $a_1, a_2, a_3, \ldots\ldots, a_n$ are n +ve (real) nos.

$$\boxed{G_n^2 = A_n \cdot H_n}$$

$$G_n = (a_1 a_2 a_3 \cdots\cdots a_n)^{\frac{1}{n}}$$

$$A_n = \frac{a_1 + a_2 + a_3 + \cdots\cdots + a_n}{n}$$

$$H_n = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \cdots\cdots + \frac{1}{a_n}}$$

c)

$$\boxed{A_n \geq G_n \geq H_n}$$

[NOTE]- equality holds if $a_1 = a_2 = a_3 = a_4 = \cdots\cdots = a_n$

- 3 nos are in A P $\rightarrow$ 1, 2, 3
- " " " " G P $\rightarrow$ 1, 2, 4
- " " " " H P $\rightarrow$ 2, 3, 6
- ★ 4 " " " " H P $\rightarrow$ 3, 4, 6, 12
- If $a^2, b^2, c^2 \Rightarrow$ A.P. so 1, 25, 49
  $|a| = 1, |b| = 5, |c| = 7$

- If $a, b, c \rightarrow AP \Rightarrow \frac{a-b}{b-c} = \frac{a}{a}$

- If $a, b, c \rightarrow$ G.P. $\Rightarrow \frac{a-b}{b-c} = \frac{a}{b}$

- If $a, b, c \rightarrow$ H.P $\Rightarrow \frac{a-b}{b-c} = \frac{a}{c}$

- $\sum ab = \frac{(\sum a)^2 - \sum (a^2)}{2}$

- $a^2, b^2, c^2 \rightarrow$ AP so $\frac{1}{b+c}, \frac{1}{a+c}, \frac{1}{a+b} \rightarrow$ AP

AGP

✓ $a, (a+d)r, (a+2d)r^2, (a+3d)r^3, \ldots\ldots$
$\ldots\ldots, (a+(n-1)d)r^{n-1}$

✓. $T_n = (a + (n-1)d)r^{n-1}$

✗. $S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2}$
$- \left(\frac{(a + (n-1)d)\ r^n}{(1-r)} \right)$

✓. $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$, if $0 < |r| < 1$
where $n \rightarrow \infty$

Sigma Notation

i) $\sum_{r=1}^{n} (a_r \pm b_r) = \sum_{r=1}^{n} a_r \pm \sum_{r=1}^{n} b_r$

ii) $\sum_{r=1}^{n} K \cdot a_r = K \cdot \sum_{r=1}^{n} a_r$

iii) $\sum_{r=1}^{n} K = K \cdot \sum_{r=1}^{n} 1 = K \cdot n$

iv) $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$

v) $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$

vi) $\sum_{r=1}^{n} r^3 = \left(\frac{n(n+1)}{2}\right)^2 = \left(\sum_{r=1}^{n} r\right)^2$

odd vii) $\sum_{r=1}^{n} (2r-1) = n^2$

even viii) $\sum_{r=1}^{n} (2r) = n(n+1)$

ix) $S_n = \sum_{r=1}^{n} T_n$

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Ch- Fundamental and Log

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YOUVA

- Some results of algebra-

1) $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ac) = \frac{1}{2}(a+b+c)((a-b)^2+(b-c)^2+(c-a)^2)$

So If $a+b+c=3abc$
then $a+b+c=0$ or $a=b=c$

2) $a^4+a^2+1 = (a^2-a+1)(a^2+a+1)$

3) Product of any three consecutive nos. is divisible by 6
$(q-1)q(q+1) = 6p$
$q \geq 2, p \geq 1$

4) $|a \pm b| \leq |a|+|b|$ iff $ab \geq 0$

5) $||a|-|b|| \leq |a \pm b|$ iff $ab \geq 0$

6) ~~$||a|-|b|| = |a-b|$ iff $ab \geq 0$~~

7) $|a-b| = |b-a|$

8) $|x-y| = |a-b| \Rightarrow (x-y) = \pm(a-b)$
or $(a-b) = \pm(x-y)$

9) $\alpha^3+\beta^3+\gamma^3 = 3\alpha\beta\gamma + (\alpha+\beta+\gamma)\{(\Sigma\alpha)^2 - 3(\Sigma\alpha\beta$

10) $a^3+b^3 = (a+b)(a^2-ab+b^2)$

11) $a^3-b^3 = (a-b)(a^2+ab+b^2)$

Log

1) $\left| \log_b^a + \log_a^b \right| \geq 2$, $\forall$ $a > 0, b > 0$, $a \neq 1, b \neq 1$

2) $\log_2 \log_2 \underbrace{\sqrt{\sqrt{\sqrt{\sqrt{\cdots\cdots\sqrt{2}}}}}}_{n \text{ times}} = -n$

3) $a^{\sqrt{\log_a^b}} = b^{\sqrt{\log_b^a}}$

4) If $x = \log_c^b + \log_b^c$

$y = \log_a^c + \log_c^a$

$z = \log_a^b + \log_b^a$

then $\boxed{x^2 + y^2 + z^2 - 4 = xyz}$

5) Constant Base

i) $\log_a^x > \log_a^y \Rightarrow \begin{cases} x > y > 0, & a > 1 \\ 0 < x < y, & 0 < a < 1 \end{cases}$

ii) $\log_a^x < \log_a^y \Rightarrow \begin{cases} 0 < x < y, & a > 1 \\ x > y > 0, & 0 < a < 1 \end{cases}$

iii) $\log_a^x > p \Rightarrow \begin{cases} x > a^p, & a > 1 \\ x < a^p, & a \in (0,1) \end{cases}$

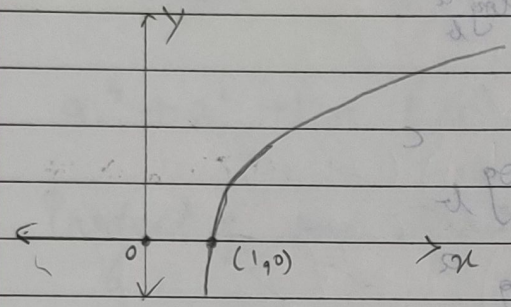
iv) $\log_a^x < p \Rightarrow \begin{cases} x < a^p, & a > 1 \\ x > a^p, & x \in (0,1) \end{cases}$

6) Variable Base

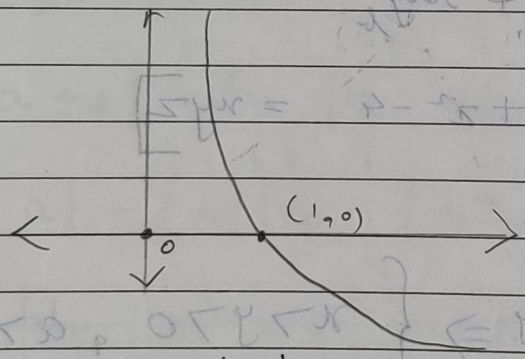
i) $\log_a x$ only if $x > 0$, $a \neq 1$, $a > 0$

7) Graph of $\log_a x$

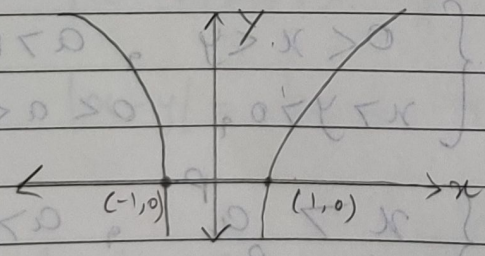
i) $y = \log_a x$, $x > 0$ & $a > 1$, $a \neq 1$



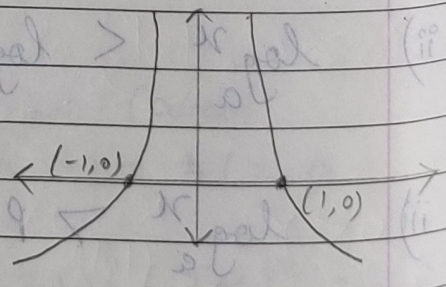
ii) $y = \log_a x$, $x > 0$, $0 < a < 1$



iii) $y = \log_a |x|$

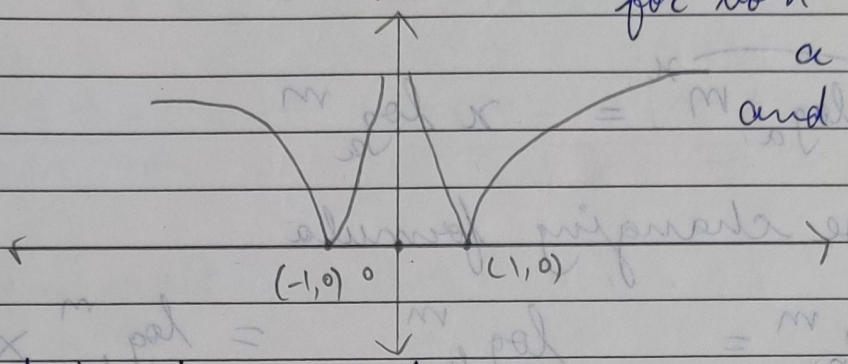


$a > 1$

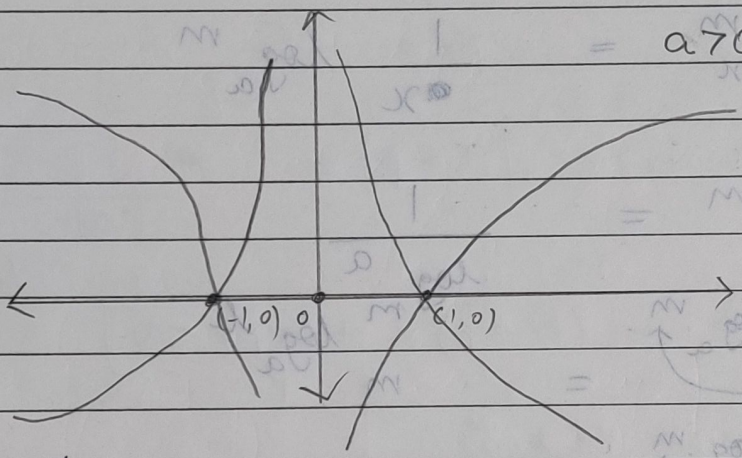


$0 < a < 1$

iv) $y = |\log_a |x||$



v) $|y| = |\log_a |x||$



8) $a^x = N \iff x = \log_a N$

$\log_a N$ is defined only when $a \neq 1$, $a > 0$, $N > 0$

i) Log of 0, -ve reals does not ~~exist~~ exist.

9) Properties of Log

a) $\log_N N = 1$

b) $\log_a m \cdot n = \log_a m + \log_a n$; $(m, n > 0)$

c) $\log_a 1 = 0$, $a > 0$, $a \neq 1$

d) $\log_a \frac{m}{n} = \log_a m - \log_a n$

e) $\log_a m^x = x \log_a m$

f) Base changing formula

$$\log_a m = \frac{\log_b m}{\log_b a} = \log_b m \times \log_a b$$

g) $\log_{a^x} m = \frac{1}{x} \log_a m$

h) $\log_a m = \frac{1}{\log_m a}$

i) $b^{\log_a m} = m^{\log_a b}$

j) $a^{\log_a m} = m$

k) $\log_{a^x} m^n = \frac{n}{x} \log_a m$

★ l) $\log_a x^{2k} = 2k \log_a |x|$, $a > 0$, $a \neq 1$, $k \in I$ (mod)

★ m) $\log_{a^{2k}} x = \frac{1}{2k} \log_{|a|} x$, $x > 0$, $a > 0$, $a \neq \pm 1$, $k \in I - \{0\}$

n) $\ln e^{a^x} = a^x$

a) $\log_a x^2 \neq 2\log_a x$, $a > 0, a \neq 1$

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Ch - Quadratic Eqⁿ

- $ax^2 + bx + c = 0$, $a \neq 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{D}}{2a}$$

$D = b^2 - 4ac$ called discriminant

1) Sum $\alpha + \beta = \frac{-b}{a}$

2) product $\alpha\beta = \frac{c}{a}$

3) diffference $|\alpha - \beta| = \frac{\sqrt{D}}{|a|}$

- $k\left[x^2 - (\alpha+\beta)x + (\alpha\beta) = 0\right]$

- ~~If~~ Nature of roots

$ax^2 + bx + c = 0$

1) $a, b, c \in R$ (real)

i) $D > 0 \Rightarrow$ real and distinct

ii) $D = 0 \Rightarrow$ Real & equal i.e. $\frac{-b}{2a} = \alpha = \beta$

iii) $D < 0 \Rightarrow$ Imaginary and conjugate of each other
$p + iq$, $p - iq$

2) $a, b, c \in Q$ (rational)

i) $D > 0$ & perfect square $\Rightarrow$ Rational

(3) Integral roots $\Rightarrow a=1,\ b,c \in I,\ D$ is a perfect square of integer

ii) $D>0$ and not perfect square $\Rightarrow$ irrational

- Relation b/t^n roots and coefficient of q.e.

$$f(x) = ax^2+bx+c \longrightarrow (1)$$

1) If one root is "0"
$\alpha=0$
$\beta = -\frac{b}{a}$ and $c=0,\ a\neq 0$

2) Both roots are zero
$b=c=0$ and $a\neq 0$

★ 3) one root ∞
$a=0$

★ 4) Both roots $\infty \Rightarrow$ $[a=b=0]$ but $c\neq 0$

5) $a=c$
$\alpha = \frac{1}{\beta}$ roots are reciprocal of one another

6) Both roots are +ve

$S = \alpha+\beta = -\frac{b}{a} = +$

$P = \alpha\beta = \frac{c}{a} = +$

$a>0,\ b<0,\ c>0$
$a<0,\ b>0,\ c<0$

7) Both roots are -ve

$S = \alpha + \beta = \frac{-b}{a} = -$	$P = \frac{c}{a} = +$
$b > 0$	$c > 0, a > 0$
$b < 0$	$c < 0, a < 0$

8) Both are of opp sign $-, +$

$b = ?$, $\frac{c}{a} = -$

$c < 0, a > 0 \quad b = ?$

$c > 0, a < 0 \quad b = ?$

* 9) one + and greater in magnitude ⊕

$S = + = \frac{-b}{a}$	$P = -ve = \frac{c}{a}$
$b < 0$	$c < 0$
	$a > 0$

$a > 0, b < 0, c < 0$

$a < 0, b > 0, c > 0$

*10) one -ve and greater in magnitude ⊕

$S = -ve = \frac{-b}{a}$	$P = -ve = \frac{c}{a}$
$a > 0, b > 0, c < 0$	
$a < 0, b < 0, c > 0$	

11) If $a+b+c=0$
one root is 1 and other is $\frac{c}{a}$

12) Roots are equal in magnitude but of opposite sign or sum is 0

$$\alpha = -\beta \quad \text{or} \quad \alpha+\beta=0$$

$$\boxed{b=0 \ , \ a \neq 0}$$

13) If $a-b+c=0$ so one root is -1 and other is $-\frac{c}{a}$

- **Ratio of Roots**

$$ax^2+bx+c=0$$

$$\boxed{\frac{\alpha}{\beta}=\frac{m}{n} \Rightarrow \frac{b^2}{ac}=\frac{(m+n)^2}{mn}}$$

- **Common roots**

1) One root is common

$$a_1x^2+b_1x+c_1=0$$
$$a_2x^2+b_2x+c_2=0$$

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \cdot \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} = \begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}^2$$

2) Both roots are in common -

$$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$$

Transformation of roots

α, β are roots of eqⁿ $ax^2 + bx + c = 0$

new roots are γ & δ put γ & $\delta = x$ and create γ & δ in new terms of x

eg. $\alpha, \beta \Rightarrow ax^2 + bx + c = 0$

$\frac{1}{3\alpha+2}$, $\frac{1}{3\beta+2}$,

$\frac{1}{3\alpha+2} = x$

$\frac{1}{x} - 2 = 3\alpha = \frac{1-2x}{x}$

$\alpha = \frac{1-2x}{3x}$

put $x \longrightarrow \frac{1-2x}{3x}$

Graph of quadratic expression

$y = ax^2 + bx + c$, $a \neq 0$

$\left(y + \frac{D}{4a}\right) = a\left(x + \frac{b}{2a}\right)^2$

$Y = AX^2$

Vertex $\equiv \left(-\frac{b}{2a}, -\frac{D}{4a}\right)$

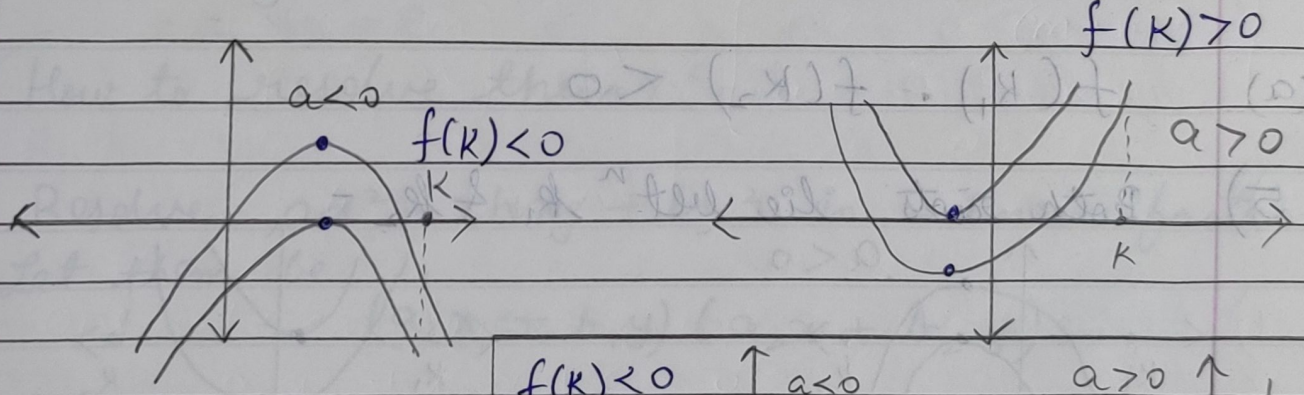
$y = ax^2 + bx + c$ @ $x = 0$, $y = c$

a decides curve up / down
b " x - coordinate
c " y - intercept
D " nature of roots (real / distinct)

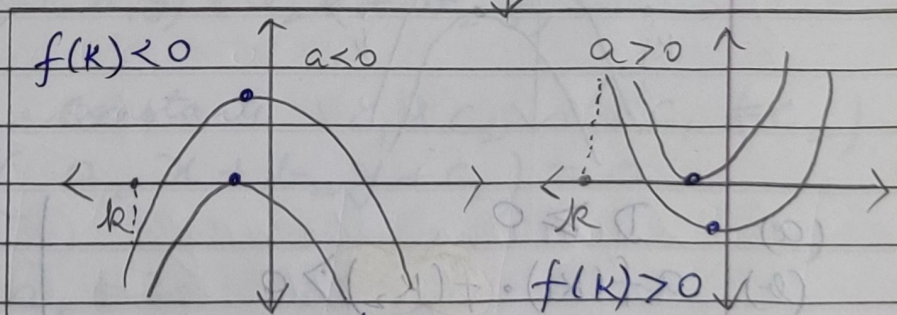
★ IMP Topic **Location of Roots**

$f(x) = ax^2 + bx + c$, $a \neq 0$
has roots α & β ($\alpha < \beta$) where $k, k_1, k_2 \in R$ ($k_1 < k_2$)

1) Both roots are less than k



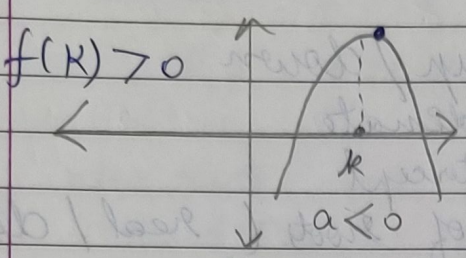
(a) $D \geq 0$
(b) $a \cdot f(k) > 0$
(c) $\frac{-b}{2a} < k$

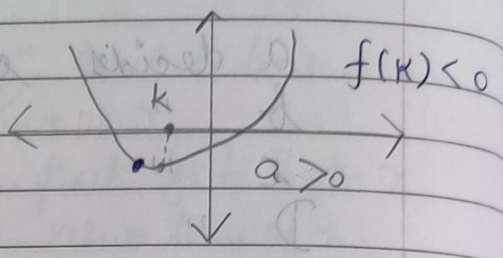


2) Both roots are greater than k -

(a) $D \geq 0$
(b) $a \cdot f(k) > 0$
(c) $\frac{-b}{2a} > k$

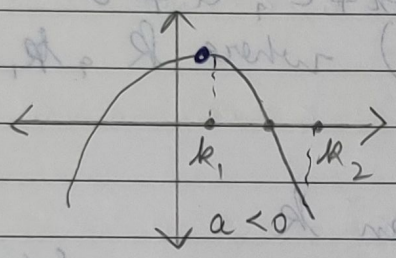
3) One root > k & another < k -

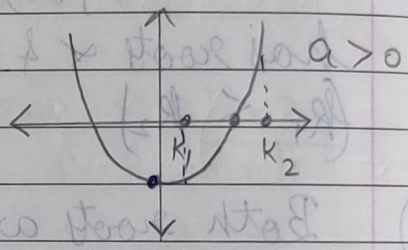




(a) $a \cdot f(k) < 0$

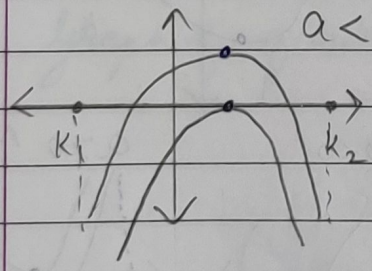
4) Exactly one root bet^n k_1 and k_2 -

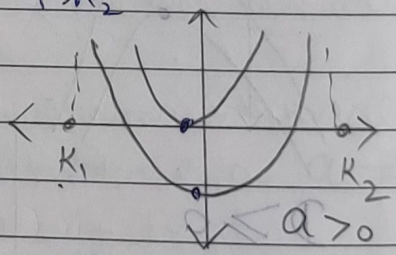




(a) $f(k_1) \cdot f(k_2) < 0$

5) Both roots lie bet^n k_1 & k_2 -





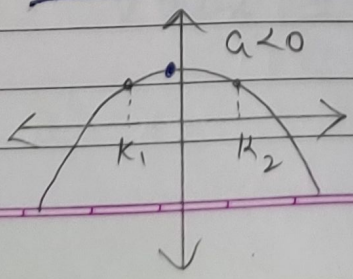
(a) $D \geq 0$

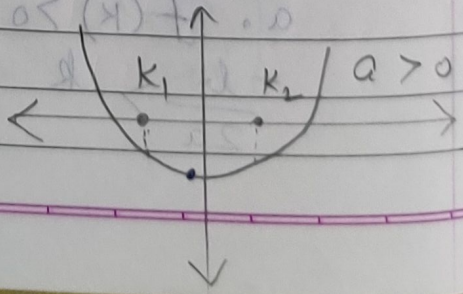
(b) $af(k_1) \cdot f(k_2) > 0$

(c) $k_1 < \frac{-b}{2a} < k_2$

(d) $a \cdot f(k_2) > 0$

6) One root < k_1 & another > k_2 -





(a) $a \cdot f(x_1) < 0$

(b) $a \cdot f(x_2) < 0$

- Condition for resolution in 2 linear factors

$$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

can be resolved in 2 linear factors iff $\Delta = 0$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

- How to resolve them

1) Resolve $ax^2 + 2hxy + by^2$ in 2 linear factors
Let them be

$$(a_1x + b_1y)(a_2x + b_2y)$$

2) Now assume two constants c_1 & c_2 $(c_1 \neq c_2)$

$$(a_1x + b_1y + c_1)(a_2x + b_2y + c_2) = 0$$

$$a_1c_2 + c_1a_2 = 2g$$

$$b_1c_2 + b_2c_1 = 2f$$

find c_1 & c_2

In this Chap. 'n' is taken as an integer if not stated otherwise

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Date:

22/3/21

Ch - Trigonometric eqⁿ

1) $\sin\theta = 0 \Rightarrow \theta = n\pi$

2) $\cos\theta = 0 \Rightarrow \theta = (2n+1)\frac{\pi}{2}$

3) $\tan\theta = 0 \Rightarrow \theta = n\pi$

4) $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n\alpha \; ; \; \alpha \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

5) $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha \; ; \; \alpha \in [0, \pi]$

6) $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha \; ; \; \alpha \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

7) $\sin^2\theta = \sin^2\alpha$
or $\cos^2\theta = \cos^2\alpha$
or $\tan^2\theta = \tan^2\alpha$
$\Rightarrow \theta = n\pi \pm \alpha$

8) $\sin\theta = 1 \Rightarrow \theta = (4n+1)\frac{\pi}{2}$

9) $\cos\theta = 1 \Rightarrow \theta = 2n\pi$

10) $\cos\theta = -1 \Rightarrow \theta = (2n+1)\pi$

- Squaring the eqⁿ @ any step should be avoided as far as possible as it leads to extra solutions.
- Avoid using formulae

$$S2x = \frac{2tx}{1+t^2}, \; C2x = \frac{1-t^2x}{1+t^2x}, \; t2x = \frac{2tx}{1-t^2x}$$

$$\text{also } \tan 3x = \frac{3tx - t3x}{1-3t^2x}$$

- Eq^n of form $a\sin\theta + b\cos\theta = c$

$$\cos(\theta - \phi) = \frac{c}{\sqrt{a^2+b^2}} \Rightarrow c(\theta - \phi) = c\alpha$$

$$\theta = 2n\pi \pm \alpha + \phi$$

If $|c| > \sqrt{a^2+b^2} \Rightarrow \theta \in \phi$ (no sol^n)

If $|c| \leq \sqrt{a^2+b^2} \Rightarrow$ So divide both sides by $\sqrt{a^2+b^2}$

- If the exponent of sines and cosines are occuring into an eq^n of even exponent then eq^n can be solved by lowering its degree.

- Eq^n of form $R(\sin x + \cos x, \sin x \cdot \cos x) = 0$

Let $\sin x + \cos x = t \Rightarrow (\sin x + \cos x)^2 = t^2$

$1 + 2\sin x \cos x = t^2 \Rightarrow \sin x \cdot \cos x = \frac{t^2 - 1}{2}$

20/7/24

Ch - Integration (O + I)

1) $\int x^n \cdot dx = \frac{x^{n+1}}{n+1} + c \quad , (n \neq -1)$

2) $\int \sin x \cdot dx = -\cos x + c$

3) $\int \cos x \cdot dx = \sin x + c$

✱4) $\int \tan x \cdot dx = \ln|\sec x| + c$

✱5) $\int \cot x \cdot dx = \ln|\sin x| + c$

✱6) $\int \sec x \cdot dx = \log|\sec x + \tan x| + c = \log\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right| + c$

✱7) $\int \operatorname{cosec} x \cdot dx = \log|\operatorname{cosec} x - \cot x| + c = \log\left|\tan\frac{x}{2}\right| +$

8) $\int \sec^2 x \, dx = \tan x + c$

9) $\int \operatorname{cosec}^2 x \, dx = -\cot x + c$

10) $\int \frac{1}{x} dx = \ln|x| + c$

11) $\int \sec x \cdot \tan x \cdot dx = \sec x + c$

12) $\int \operatorname{cosec} x \cdot \cot x \cdot dx = -\operatorname{cosec} x + c$

13) $\int e^{ax+b} = \frac{e^{ax+b}}{a} + c$

14) $\int (ax+b)^n = \frac{(ax+b)^{n+1}}{a(n+1)} + c \quad , (n \neq -1)$

15) $\int \frac{1}{ax+b} = \frac{\ln|ax+b|}{a} + c$

16) $\int a^{px+q} = \frac{a^{px+q}}{p \ln a} + c$

17) $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + c$

18) $\int \frac{dx}{|x|\sqrt{x^2-a^2}} = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + c$

mod x

19) $\int \frac{dx}{\sqrt{a^2-x^2}} = \frac{1}{1} \sin^{-1}\left(\frac{x}{a}\right) + c$

20) $\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left[x + \sqrt{x^2+a^2}\right] + c$

21) $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left[x + \sqrt{x^2-a^2}\right] + c$

22) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln\left|\frac{a+x}{a-x}\right| + c$

23) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + c$

* Some standard substitution ⇒

1) $a^2+x^2 \Rightarrow x = a\tan\theta$ or $x = a\cot\theta$

2) $a^2-x^2 \Rightarrow$ ~~$x = a\sec\theta$ or~~
$x = a\sin\theta$ or $x = a\cos\theta$

3) $x^2-a^2 \Rightarrow x = a\sec\theta$ or $x = a\operatorname{cosec}\theta$

4) $\sqrt{\frac{a-x}{a+x}} \Rightarrow x = a\cos 2\theta$

5) $\sqrt{\frac{x-\alpha}{\beta-x}}$ or $\sqrt{(x-\alpha)(\beta-x)} \Rightarrow x = \alpha c^2\theta + \beta s^2\theta$

6) $\sqrt{\frac{x-\alpha}{x-\beta}}$ or $\sqrt{(x-\alpha)(x-\beta)} \Rightarrow x = \alpha\sec^2\theta - \beta\tan^2\theta$

7) $\sqrt{(x-\alpha)(x-\beta)} \Rightarrow x-\alpha = t^2$ or $x-\beta = t^2$

* Some properties of I-I.

1) $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$

2) $\int f'(x)\,(f(x))^n = \frac{(f(x))^{n+1}}{n+1} + c$

3) $\int u\cdot v\cdot dx = u\left[\int v\,dx\right] - \int\left[\left(\frac{du}{dx}\right)\cdot\int v\,dx\right]dx$

ILATE

$\frac{d}{dx}\left(\frac{ax+b}{cx+d}\right)=\frac{ad-bc}{(cx+d)^2}$ | $\int \ln x \cdot dx = x(\ln x - 1) + C$

4) $\int e^x \left[f(x) + f'(x) \right] dx = e^x \cdot f(x) + C$

5) $\int \left[f(x) + x f'(x) \right] dx = x f(x) + C$

6) $\int e^x \left[f(x) - f''(x) \right] dx = e^x \left[f(x) - f'(x) \right] + C$

7) $\int \frac{ad-bc}{(cx+d)^2} = \frac{ax+b}{cx+d}$

- $\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2+b^2}\left(a \sin bx - b \cos bx\right) + C$

- $\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2+b^2}\left(a \cos bx + b \sin bx\right) + C$

✡ Manipulating Integrands ⇒

i) $\int \frac{dx}{x(x^n+1)}$, $n \in N$, take x^n common & put $1+x^{-n} = t$

ii) $\int \frac{dx}{x^2 (x^n+1)^{\frac{n-1}{n}}}$, put $1+x^{-n} = t^n$

iii) $\int \frac{dx}{x^n (1+x^n)^{\frac{1}{n}}}$, take x^n common & put $1+x^{-n} = t^n$

iv) $\int \frac{dx}{(ax+b)\sqrt{px+q}}$ & $\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$

put $px+q = t^2$

v) $\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$ put $ax+b = \frac{1}{t}$

vi) $\int \frac{dx}{L_1\sqrt{L}} + \int \frac{L_1}{\sqrt{L}}dx + \int \frac{\sqrt{L}}{L_1}dx$

L = linear

L_1 = " "

Substitute $L = t^2$

vii) $\int \frac{dx}{L\sqrt{Q}}$, $L = \frac{1}{t}$

Definite Integration

- $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

- $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

- $\int_{-a}^{+a} f(x)\,dx = \int_0^a [f(x) + f(-x)]\,dx = \begin{cases} 0, & f \text{ is odd} \\ 2\int_0^a f(x)\,dx & \text{if } f \text{ is even.} \end{cases}$

- $\int_a^b f(ax)\,dx = \int_a^b f(a+b-x)\,dx$

- $\int_0^{nT} f(x)\,dx = n\int_0^T f(x)\,dx$, T = period
  $f(T+x) = f(x)$
  $n \in I$

- $\int_{mT}^{nT} f(x)\,dx = (n-m)\int_0^T f(x)\,dx$, T is period of f.
  $n, m \in I$.

- $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n} \cdot f\left(\frac{r}{n}\right) = \int_0^1 f(x)\,dx$

- $\int_0^{\pi/4} \ln(1+\tan x)\,dx = \frac{\pi}{8}\ln 2$

- $\int_0^{\pi/2} \log(\sin x)\,dx = \int_0^{\pi/2} \log(\cos x) = -\frac{\pi}{2}\log 2$

* Newton - Leibnitz Rule ⇒

1) $$\frac{d}{dx}\left[\int_{g(x)}^{h(x)} f(t)\,dt\right] = f(h(x))\,h'(x) - f(x)\,g'(x)$$

2) $$\frac{d}{dx}\left[\int_{g(x)}^{h(x)} f(x,t)\,dt\right] = \left[f(x,h(x))\,h'(x) - f(x,g(x))\,g'(x)\right] + \int_{g(x)}^{h(x)} f'(x;t)\,dt.$$

$f'(x;t)$ → partially $\frac{\partial f}{\partial x}$ wrt x keeping t constant

- If $f(x)$ is continous in $[a,b]$ & its range is $[m,M]$, then
$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$$
- If $f(x) \leq \phi(x) \quad \forall \quad x \in [a,b]$
$$\int_a^b f(x)\,dx \leq \int_a^b \phi(x)\,dx$$
- $$\left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx$$
- Average of a fn over a interval $[a,b]$ is defined as
$$\langle y \rangle = \frac{1}{(b-a)} \int_a^b f(x)\,dx$$

- Root Mean Square of a fn on $[a,b]$ is

$$= \sqrt{\frac{1}{b-a}\int_a^b (f(x))^2\,dx}$$

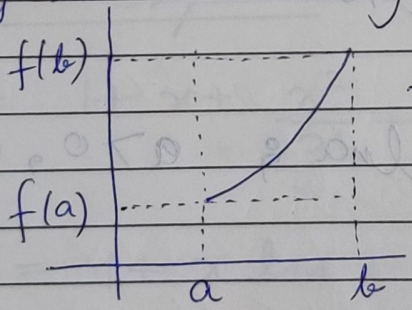
- $$\int_a^b f(x)\,dx + \int_{f(a)}^{f(b)} f^{-1}(x)\,dx = b\,f(b) - a\,f(a)$$

- $$\int_a^b f(x)\,dx + \int_\alpha^\beta g(y)\,dy = b\,f(b) - a\,f(a)$$
$$= b\beta - a\alpha$$

$g(x) = f^{-1}(x)$

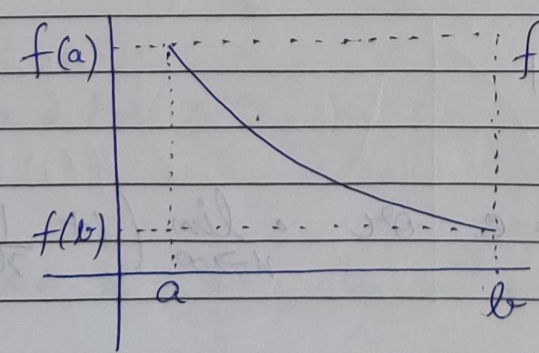
* Estimation of Definite Integration & general inequalities

1) If f is monotonically $\uparrow^{ng}$ in (a,b)



$$f(a)(b-a) < \int_a^b f(x)\,dx < f(b)(b-a)$$

2) If f is monotonically $\downarrow$ in (a,b)



$$f(b)(b-a) < \int_a^b f(x)\,dx < f(a)(b-a)$$

29/7/24

Ch - Limits

- $\lim_{x \to 0} \frac{\sin x}{x} = 1$
- $\lim_{x \to 0} \frac{\tan x}{x} = \lim_{x \to 0} \frac{x}{\tan x} = \lim_{x \to 0} \frac{x}{\sin x} = 1$
- $\lim_{x \to 0} \frac{x}{\sin^{-1} x} = \lim_{x \to 0} \frac{x}{\tan^{-1} x} = \lim_{x \to 0} \frac{\sin^{-1} x}{x} = 1 = \lim_{x \to 0} \frac{\tan^{-1} x}{x}$

★ If $\lim_{x \to a} f(x) = 0$ then $\lim_{x \to a} \frac{\sin(f(x))}{f(x)} = 1$

- $\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
- $\lim_{x \to a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1}$
- $\lim_{x \to 0} \frac{(1+x)^n - 1}{x} = n$
- $\lim_{x \to 0} \frac{a^x - 1}{x} = \ln a \; ; \; a > 0, a \neq 1$
- $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x \to 0} (1+x)^{\frac{1}{x}} = e$ or $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e$

✱ If $\lim_{x\to a} f(x) = 0$ then $\lim_{x\to a} \frac{a^{f(x)} - 1}{f(x)} = \ln a$

$a > 0, a \neq 1$

✱ If $\lim_{x\to a} f(x) = 0$ then $\lim_{x\to a} (1 + f(x))^{\frac{1}{f(x)}} = e$

✱✱ If $\lim_{x\to a} f(x) = 1$ & $\lim_{x\to a} \phi(x) = \infty$

then ; $\lim_{x\to a} [f(x)]^{\phi(x)} = e^K$

where $K = \lim_{x\to a} \phi(x)[f(x) - 1]$

✱✱ If $\lim_{x\to a} f(x) = A > 0$ & $\lim_{x\to a} \phi(x) = B$

then $\lim_{x\to a} [f(x)]^{\phi(x)} = A^B$

✱ Series expansion ⇒

i) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$

ii) $a^x = 1 + x \ln a + \frac{x^2}{2!}(\ln a)^2 + \cdots$

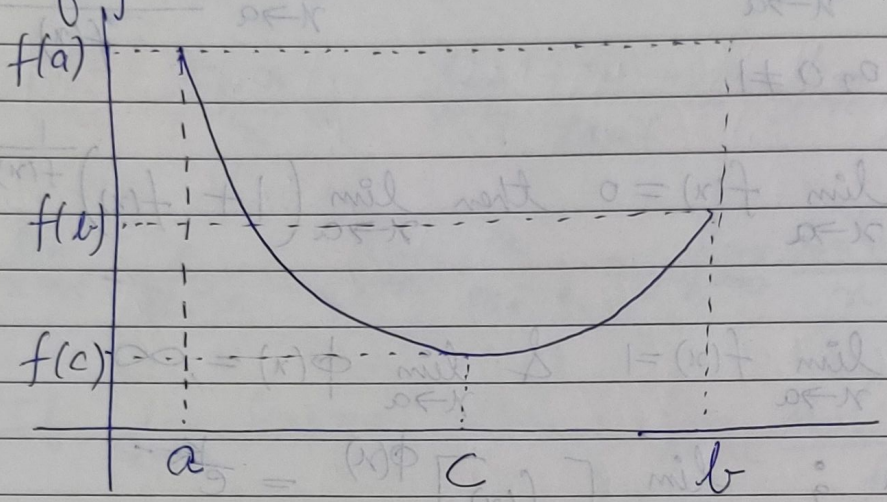
iii) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

iv) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

v) $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \cdots$

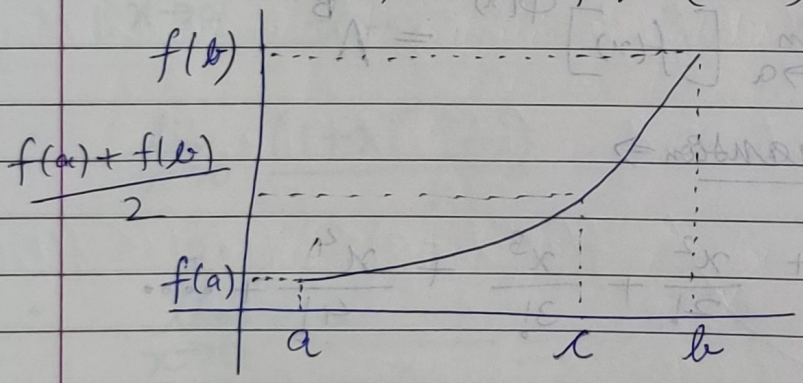
graph of e^{-x^2} (even fn)

3) If f is non monotonic in (a,b), $c \in (a,b)$



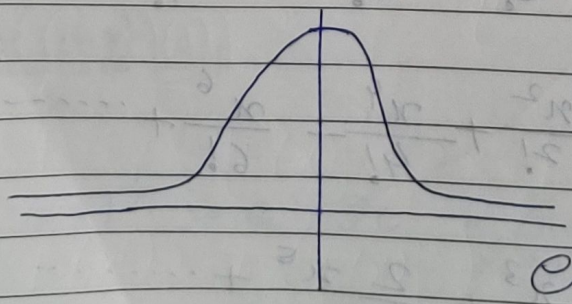
$$f(c)(b-a) < \int_a^b f(x)\,dx < f(a)(b-a)$$

4) • If fn $f(x) \uparrow$ & has concave up graph in $[a,b]$ i.e. $f'(x)$ & $f''(x) > 0$ (both) then



$$(b-a)\,f(a) < \int_a^b f(x)\,dx < (b-a)\left(\frac{f(a)+f(b)}{2}\right)$$

•



graph of $f(x) = e^{-x^2}$ even fn.

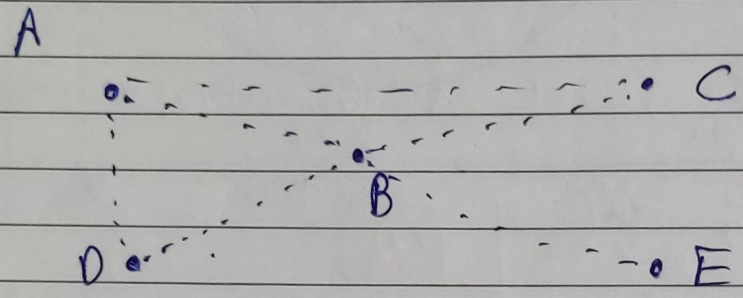
- If $f(x)$ & $g(x)$ are integrable on interval (a, b)

$$\left| \int_a^b f(x) \cdot g(x) \cdot dx \right| \leq \sqrt{\left(\int_a^b f^2(x) dx \right) \left(\int_a^b g^2(x) \cdot dx \right)}$$

Om Namoh Bhagwate Vasudevaay Namah

7/1/25 – Tuesday – 2:45 PM – Mathematics.

* **Markov Chains**



Property: Given the present, the future is independent of the past.

→ Probability distribution stabilises after a certain point (higher number of events eg 1000, 1500)

Expectation $E(X) = \sum_{x \in X} x \cdot P(x)$

Variance $V(X) = E\left[(X - E(X))^2\right]$

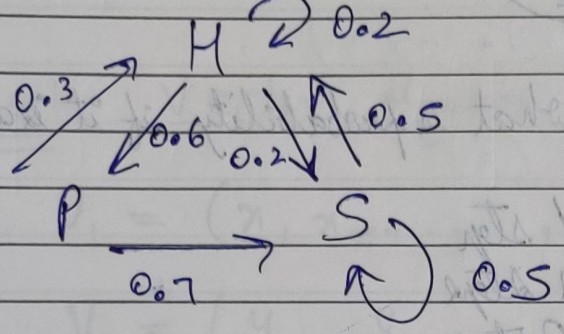
$$V(X) = E(x^2) - \left[E(x)\right]^2$$

$$V(X) = E\left[X^2 + \left[E(x)\right]^2 - 2x \cdot E(x)\right]$$

$$E(X^2) + E\left(\left[E(x)\right]^2\right) - 2\,E(x) \cdot E(E(x))$$

- Stationary distribution / Equilibrium state.

Q)



$$A = \begin{array}{c} \\ H \\ P \\ S \end{array} \begin{array}{c} \begin{array}{ccc} H & P & S \end{array} \\ \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix} \end{array} \begin{array}{l} \Rightarrow 1 \\ \Rightarrow 1 \\ \Rightarrow 1 \end{array} \Big\} \Sigma$$

$$\pi_0 = [0 \;\; 1 \;\; 0]$$

$$\pi_0 \cdot A = [0\,1\,0] \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix} = [0.3 \;\; 0 \;\; 0.7]$$

$$\pi_1 A = [0.3 \;\; 0 \;\; 0.7] \; A = [0.41 \;\; 0.18 \;\; 0.41]$$

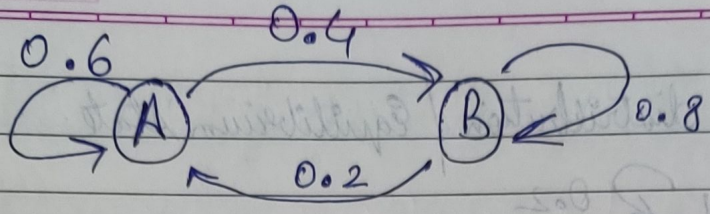
$\pi A = \pi$ → Stationary state. $\lambda = 1$

π = Left Eigen Vector of A

$Av = \lambda v$ → Eigen Vector Eq

$\pi[1] + \pi[2] + \pi[3] = 1$ → $\pi = [0.35211 \;\; 0.21127 \;\; 0.43662]$

Q)



If it starts at A, what is probability if it is at A & B after

i) 1 step
ii) n steps
iii) ∞ steps.

Ans)

	A	B
A	0.6	0.2
B	0.4	0.8

$$P_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \leftarrow A \\ \leftarrow B$$

i) $P_1 = A \cdot P_0 = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}$

ii) n-steps

$P_1 = A \cdot P_0$, $P_2 = AP_1 = A^2 P_0$

$$P_n = A^n \cdot P_0$$

$A = U D U^{-1}$

$\lambda = 1$ $\quad U_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\lambda = 0.4 \qquad u_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Lecture-1 Vector Algebra

$$V_1 = (x_1, x_2, \ldots, x_n) \in R^n$$

$$V_2 = (y_1, y_2, \ldots, y_n) \in R^n$$

$$V_1 \cdot V_2 = x_1 y_1 + x_2 y_2 + \ldots + x_n y_n$$

$$= \sum_{i=1}^{n} x_i \cdot y_i$$

Length

$$\vec{V} = (x_1, x_2, x_3, \ldots, x_n)$$

$$|\vec{V}| = V = \sqrt{\vec{V} \cdot \vec{V}} = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}$$

$$|\vec{V}| = \sqrt{\sum_{i=1}^{n} x_i^2}$$

Angle

$$V_1, V_2 \in R^n$$

$$\theta = \cos^{-1}\left(\frac{\vec{V_1} \cdot \vec{V_2}}{|\vec{V_1}| \cdot |\vec{V_2}|} \right)$$

* Properties of expectations

1) $E(X+Y) = E(X) + E(Y)$

2) $E(c \cdot X) = c \cdot E(X)$

3) $E(XY) = E(X) \cdot E(Y)$

Q) $\Omega = \{1, 2, 3, 4, 5, 6\}$

Variance = ? of a die roll

$$E(X) = \sum_{x=1}^{6} X \cdot P(X)$$

$$= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$

$$E(X) = \frac{21}{6} = 3.5$$

$$E(X^2) = \sum_{x=1}^{6} X^2 \cdot P(X)$$

$$= 1 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 9 \cdot \frac{1}{6} + 16 \cdot \frac{1}{6} + 25 \cdot \frac{1}{6} + 36 \cdot \frac{1}{6}$$

$$= \left(\sum_{k=1}^{6} k^2 \right) \cdot \frac{1}{6}$$

$$= \frac{6(7)(13)}{6} \times \frac{1}{6} = \frac{91}{6}$$

$$Var(X) = E(X^2) - [E(X)]^2$$

$$= \frac{91}{6} - \left(\frac{7}{2}\right)^2$$

$$= \frac{91}{6} - \frac{49}{4} = \frac{182 - 147}{12}$$

$$= \frac{35}{12} \approx \boxed{2.92}$$

$Var = \sigma^2$ (Standard deviation)

* **Markovs Inequality**

$$P(X \geq a) \leq \frac{E(X)}{a}$$

In Measure theory or Chebyshev's Inequality

$$\rightarrow \mu(\{x \in X : |f(x)| \geq \varepsilon\}) \leq \frac{1}{\varepsilon} \int_X |f| \, d\mu.$$

→ If $X \geq 0$ and $E(X)$ is small, then X is unlikely to be very large.
$a \in$ constant (+ve)

10 December 2025 - Wednesday - 8:50PM

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Ch - Differentiation

i) $\frac{d}{dx}(\text{constant}) = 0$

ii) $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$

iii) $\frac{d}{dx}(|x|) = \frac{x}{|x|}$

iv) $\frac{d}{dx}(\ln|x|) = \frac{1}{x}$

v) $\frac{d}{dx}(e^x) = e^x$

vi) $\frac{d}{dx}(a^x) = a^x \cdot \log a$

vii) $\frac{d}{dx}(\log_a|x|) = \frac{1}{x \cdot \log_e a}$

viii) $\frac{d}{dx}(\sin x) = \cos x$

ix) $\frac{d}{dx}(\cos x) = -\sin x$

x) $\frac{d}{dx}(\tan x) = \sec^2 x$

xi) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

xii) $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$

xiii) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$

xiv) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$, $|x| < 1$

xv) $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$, $-1 < x < +1$

xvi) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$, $x \in R$

xvii) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$, $|x| > 1$ or $x \in R - [-1, 1]$

xviii) $\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$, $x \in R$

xix) $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}$, $|x| > 1$ or $x \in R - [-1, 1]$

✳ Rules of differentiation

1) $y = u(x) \pm v(x) \pm w(x) \pm \cdots$

$$\frac{dy}{dx} = \frac{du(x)}{dx} \pm \frac{dv(x)}{dx} \pm \frac{dw(x)}{dx} \pm \cdots$$

2) $y = k \cdot f(x)$; k = constant

$$\frac{dy}{dx} = k \cdot \frac{df(x)}{dx}$$

✳ Product Rule

3) $y = u(x) \cdot v(x)$

$$\frac{dy}{dx} = \left\{\frac{du(x)}{dx}\right\} \cdot v(x) + u(x) \cdot \left\{\frac{dv(x)}{dx}\right\}$$

✳ Quotient Rule

4) $y = \frac{u(x)}{v(x)}$

$$\frac{dy}{dx} = \frac{\left\{\frac{du(x)}{dx}\right\} \cdot v(x) - \left\{\frac{dv(x)}{dx}\right\} \cdot u(x)}{\{v(x)\}^2}$$

5) Chain Rule

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$